3/10/24

classmate
Date
Page

Experiment - 14

Aim:

Write a Code Using Raw Sockets to implement packet Sniffing

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP
def packet_callback(packet):
    if ip in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown protocol"
        print(f"protocol: {protocol_name}")
        print(f"Destination IP: {dst_ip}")
        print("_" * 40)
sniff(iface="Wifi", prn=packet_callback,
      filter="ip", store=0)
```

8)

```
Protocol: TCP
Source IP: 57.155.141.117
Destination IP: 192.168.1.6

Protocol: TCP
Source IP: 192.168.1.6
Destination IP: 57.155.141.117

Protocol: TCP
Source IP: 192.168.1.6
Destination IP: 57.155.141.117

Protocol: TCP
Source IP: 57.155.141.117
Destination IP: 192.168.1.6

Protocol: TCP
Source IP: 57.155.141.117
Destination IP: 192.168.1.6

Protocol: UDP
Source IP: 192.168.1.6
Destination IP: 192.168.1.1

Protocol: UDP
Source IP: 192.168.1.6
Destination IP: 192.168.1.1

Protocol: UDP
Source IP: 192.168.1.6
Destination IP: 192.168.1.1

Protocol: UDP
Source IP: 192.168.1.1
Destination IP: 192.168.1.6
```

Result:

program to implement packet sniffing has been executed successfully.